

backend/api/chat.py Documentation (Roman Urdu)

Yeh AI web app ke liy sb sy ziada invoke (chalnay walii) file hai. Job bb user textbox ma msg type karta hy, request is script ki endpoints py strike hoti hy.

File Ka Maqsad (Purpose)

Iska basic purpose User Request receive krna, us par context aur memory checks (Distillation engine p phelna), LLM sy jawab extract kr k wapis web page par as a stream bhejna (typewriter effect laaney kliy) ha.

Code Blocks aur Endpoints ki Tafseel

1. POST /chat/web-search

- **Kya karta hai?:** Yeh specifically un queries k liy kam ata h jb user nekt "Web Search" toggle on kya hua ho. Yeh route direct DuckDuckGo py search hit krta a, top 3 results ka HTML code scrap karta ha aur LLM ko summarize k liy dai dta ha.

2. POST /chat (Ahem tareen function)

Logic ar Flow:

1. Check karta ha system m Ollama Zinda ha aur selected model mojood ha ya nii.
2. Check krta ha purani chat hai ya nai conversation. Agar nai tou Database ma nai Thread entry marta ha!
3. `classify_intent()` function call karta ha k is sawal ko purane kitne message chahiye (context striction).
4. Subsay bari cheez: Yeh check karta ha k User nai Document attach kia hua ha thread mein? Agr kia hy or RAG on ha, tou `rag_pipeline.distilled_query` route sy PDFs dhoond kar lata h.
5. Jawab return hone se phelay background k andr data base k andr metric (compression ratio k results) store karadeta ha!
6. System prompt apply kar k `_stream_chat` run kar dyta hi.

3. POST /chat/continue

- **Kya karta hai?:** Agar LLM likhty likhty lambai (token capacity) ki base p beech mn "Stop" hojay, user UI par "**Continue Generation**" p push karta ha. Yeh point dobara Ollama ko command dy kr adhoora kam mukammal karwata hv.

4. `_stream_chat(generator function)`

- **Kya karta hai?:** Chunk by Chunk alphaz web pe pass krata ha over HTTP SSE (Server-Sent Events) ar sath m `TokenTracker` me hisaab b banata rehta k tokens zyda ho re ya nai tak continuation limits check ho skin.